

Practice Questions Sheet 1

This sheet provides practice with simple linear regression. You will work with the regression model algebraically, fit and interpret a model in **R**, and reproduce key regression calculations by hand. These practice questions should be attempted in your own time. These questions are **not graded** and **should not be handed in**

You should attend the joint ST302/ST303 lab to get help with the questions and have your answers checked.

Student name: _____

Student number: _____

Simple linear regression

Consider the following simple linear regression model

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, \dots, n.$$

The above model may be alternatively written as

$$y_i = \alpha_0 + \alpha_1(x_i - \bar{x}) + \epsilon_i, \quad i = 1, \dots, n.$$

Answer the following:

1. Using the method of least squares, derive expressions for the estimators $\hat{\alpha}_0$ and $\hat{\alpha}_1$
2. Interpret the parameters α_0 and α_1 .
3. Explain why mean-centering the predictor variable can be useful.

Simple linear regression in R

As concrete cures, it gains strength overtime. The following data (see Table 1) represent the 7-day and 28-day strength, in pounds per square inch (psi), of a certain type of concrete.

Table 1: Concrete data

7 Day	28 Day
2300	4370
2430	4640
2890	4620
3120	4900
3380	5020
3390	5220

You can read the data from the RStudio server using:

```
# Using the `readr` and the `here` package.  
# TIP: you can press tab when writing the folder path to have all the options  
# folder / file options appear to you  
  
concrete <- readr::read_csv(here::here("Sharedfiles/ST303/data/Concrete.csv"))
```

Answer the following set of questions using R

1. Create a scatterplot showing the relationship between 7-day strength on the x -axis and 28-day strength on the y -axis.
2. Does it seem appropriate to assume a linear relationship between the two variables (explain why you think so)?
3. Specify a simple linear regression model for these data algebraically.
4. Fit this model in R and report what the **intercept**, **slope**, and **standard error** for each term are.
5. In the context of this data, what do these values mean?
6. Update the scatter plot you made earlier to include a fitted line.

Simple linear regression by hand

This time, instead of using R, we will be calculating our linear regression parameters by hand. Using the concrete data we mentioned in Table 1, calculate the following:

1. $\bar{x}$, $\sum_{i=1}^n x_i^2$, and S_{xx} .
2. $\bar{y}$, $\sum_{i=1}^n y_i^2$, and S_{yy} .
3. $\sum_{i=1}^n x_i y_i$ and S_{xy} .

You may use R to verify your calculations.

However, in an exam setting you may be asked to do calculations by hand (*hint, hint, wink, wink!!*).

As such, please show all of your workings.

Use your calculated quantities to:

4. Calculate the least-squares slope estimate and the least-squares intercept estimate
5. In addition, calculate the estimated error variance.